StarryOS

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StarryOS contributors

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StarryOS Rust Linux `Cargo.toml` “A Linux-compatible OS kernel built on ArceOS unikernel” crate `starry-kernel` `starryos` init

ArceOS/ax Linux `kernel/Cargo.toml` `axruntime` `axhal` `axtask` `axmm` `axfs` `axnet` `uspace` `multitask` `sched-rr` `fs-ext4` `fs-fat` `rtc` feature ELF / Linux

init	`src/main.rs` init `kernel/src/entry.rs` init ELF
	`kernel/src/syscall/mod.rs` `syscalls::Sysno` fs mm task signal net ipc sync time io_mpx resources
	`AddrSpace` `MemorySet<Backend>` alloc CoW shared file linear
/	`Thread` `axtask` `ProcessData` futex rlimit
	`FileLike` socket eventfd timerfd signalfd pidfd BPF `/dev` `/dev/shm` `/tmp` `/proc` `/sys`
	`starry_signal` `rt_sigaction` `rt_sigprocmask` `kill/tkill/tgkill` job-control stop/continue
	`net` feature socket `axnet` TCP/UDP Unix socket send/recv socket options `ENOSYS`
IPC/	futex robust list SysV message queue SysV shared memory eventfd pipe epoll/poll/select
	`/dev/fb0` `axdisplay::has_display()` `input` feature evdev tty/pty rtc loop random zero null

“ ”

X11/Wayland eBPF verifier Linux VM

	`Cargo.toml`	`workspace.package.description` Linux-compatible ArceOS unikernel
. . .	`kernel/src/lib.rs`	`pub mod entry` `config/file/mm/pseudofs/syscall/task/time`
. . 开关	`Cargo.toml`, `kernel/Cargo.toml`	`qemu` `starry-kernel/net` feature `net/input/dev-log/memtrack`

StarryOS

	`src/main.rs`, `src/init.sh`	init BusyBox shell LTP `starry_kernel::entry::init`
	`kernel/src/entry.rs`	init ELF stdio init
Linux	`kernel/src/syscall`, `kernel/src/task`, `kernel/src/mm`, `kernel/src/file`, `kernel/src/pseudofs`	Linux
ArceOS/ax	`axruntime`, `axhal`, `axtask`, `axmm`, `axfs`, `axnet`	

`handle_syscall` `Sysno`
socket eventfd `FileLike`
`UserPtr` `UserConstPtr` `starry_vm`

. . .

```
  -> UserContext::run()
  -> syscall / page fault / exception / interrupt
  -> syscall::{fs, mm, task, signal, net, ipc, sync, time, io_mpx}
  -> task::ProcessData / mm::AddrSpace / file::FileLike / pseudofs
  -> axfs / axnet / axtask / axhal / axmm
```

`Thread` `axtask::TaskInner` `axtask` /
`scope-local` `starry_process::Process` OS
`ProcessData`

`src/main.rs`	`starryos` init feature shell/debug
`kernel/src/entry.rs`	initproc
`kernel/src/syscall`	Linux
`kernel/src/mm`	ELF mmap/brk
`kernel/src/task`	futex timer
`kernel/src/file`	`FileLike` pipe eventfd timerfd signalfd epoll pidfd bpf socket
`kernel/src/pseudofs`	devfs procfs tmpfs SimpleFs/Device
`kernel/src/config`	trampoline

Feature

`qemu` feature PCI `vf2` VisionFive2 SDMMC `smp` axfeat SMP `shell-init` shell crate `net` `input` `dev-log` `memtrack` feature `kernel/src/syscall/net` `#[cfg(feature = "net")]` `/dev/input` `#[cfg(feature = "input")]`

. . .	`kernel/src/lib.rs`	`entry/file/mm/pseudofs/syscall/task/time`
. .	`kernel/src/task/mod.rs`	`Thread` `TaskExt` `AsThread` `TaskInner`
. . . .	`kernel/src/file/mod.rs`	`FileLike` read/write/stat/ioctl/poll/path fd `FlattenObjects<FileDescriptor, AX_FILE_LIMIT>`
. .	`kernel/src/syscall/net/mod.rs`	`net` `sys_socket` `ENOSYS` re-export net

`src/main.rs` `starryos` feature

`timerfd-debug-init`	`include_str!("init_timerfd_debug.sh")`	timerfd `src/`

`shell-init`	`src/init_shell.sh`	shell `STARRY_SHELL_MODE=1`
	`src/init.sh`	rootfs

`["/musl/busybox", "sh", "-c", INIT_SCRIPT]` `src/main.rs` `src/ltp_testcases.sh` `LTP_CASES` init LTP

init

`kernel/src/entry.rs::init` init

```
pseudofs::mount_all()
spawn_alarm_task()
resolve_init_command()
new_user_aspace_empty()
copy_from_kernel()
load_user_app()
UserContext::new(entry, stack, 0)
new_user_task()
Process::new_init()
N_TTY.bind_to()
ProcessData::new()
add_stdio()
spawn_task()
task.join()
unmount_all() / flush() / system_off()
```

StarryOS ELF shell `resolve_init_command` BusyBox fallback `/musl/busybox` `/glibc/busybox`

KERNEL_STACK_SIZE	0x4_0000	TaskInner::new
USER_SPACE_BASE	0x1000	
USER_STACK_TOP	0x4_0000_0000 0x7fff_0000_0000	
USER_HEAP_BASE	0x4000_0000	brk
USER_INTERP_BASE	0x400_0000	
SIGNAL_TRAMPOLINE	0x6000_1000	trampoline

feature

. . . .	src/main.rs	CMDLINE BusyBox shell init starry_kernel::entry::init
init	kernel/src/entry.rs	mount_all load_user_app new_user_task Process::new_init add_stdio spawn_task
. . . .	src/init.sh	/bin wrapper
. . .	kernel/src/config/*.rs	target_arch signal trampoline

kernel/src/syscall/mod.rs::handle_syscall UserContext
syscalls::Sysno::new -ENOSYS
match sys_* AxResult<isize> Linux

handle_syscall Sysno
x86_64 open mkdir poll/select ABI
openat mkdirat

/	syscall/fs	open/read/write/stat/mount/xattr/aio/memfd/ inotify/fanotify/pidfd/signalfd/pipe/eventfd
I/O	syscall/io_mpx	epoll poll ppoll select/pselect

	`syscall/mm`	brk mmap munmap mprotect mremap madvise msync mlock mincore
	`syscall/task`	clone/clone3 execve exit wait sched ptrace
	`syscall/signal.rs`	rt_sigaction rt_sigprocmask kill tkill tgkill sigpending sigsuspend
	`syscall/net`	socket bind/connect/listen/accept send/recv getsockopt/setsockopt `net` feature
IPC	`syscall/ipc`	SysV message queue shared memory
	`syscall/sync`	futex robust list membarrier
/ /	`syscall/sys.rs`, `resources.rs`, `time.rs`	uname prlimit getrlimit clock/gettimeofday/nanosleep/timer
BPF	`syscall/bpf.rs`, `file/bpf.rs`	map/prog fd eBPF

`mm::UserPtr<T>` `mm::UserConstPtr<T>` `starry_vm::VmPtr` `UserPtr` `populate_area` null-terminated buffer `VmBytes/VmBytesMut`

`axerrno::{AxError, LinuxError}` `AxResult<isize>` `SYSCALL_RESTART`

BPF

`sys_bpf` `BPF_MAP_CREATE` `BPF_MAP_LOOKUP_ELEM` `BPF_MAP_UPDATE_ELEM` `BPF_PROG_LOAD` `file/bpf.rs` map hash array ringbuf program `Prog01` `Prog02` `Prog05` `Noop` `SO_ATTACH_BPF` verifier JIT eBPF

. . .	`kernel/src/syscall/mod.rs`	`handle_syscall` `Sysno` fs/mm/task/net/signal/time/ipc/sync
. .	`kernel/src/mm/access.rs`	`UserPtr/UserConstPtr` `check_region` `can_access_range` `populate_area`
.	`kernel/src/syscall/net/mod.rs`	`#[cfg(not(feature = "net"))]` socket `ENOSYS`
`BPF` .	`kernel/src/syscall/bpf.rs`, `kernel/src/file/bpf.rs`	map / / program load attach

`mm::AddrSpace`

`va_range`	`VirtAddrRange`	
`areas`	`MemorySet<Backend>`	
`pt`	`PageTable`	map/unmap/protect/query/clone

`kernel/src/config/*.rs` riscv64/loongarch64 `0x1000` `0x4_0000_0000` x86_64/aarch64 `0x7fff_0000_0000` `0x4000_0000` `brk` `USER_HEAP_SIZE_MAX`

`AddrSpace` `Backend`

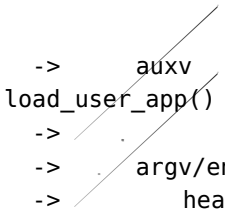
`Linear`	`backend/linear.rs`	mmap framebuffer
`Cow`	`backend/cow.rs`	fork/clone frame refcount
`Shared`	`backend/shared.rs`	shared mmap SysV shared memory
`File`	`backend/file.rs`	mmap `CachedFile` readahead evict listener

`BackendOps::populate` VMA populate / `mmap` `MAP_POPULATE` populate

ELF

`kernel/src/mm/loader.rs`

```
resolve_exec_path()
ElfLoader::load()
  ->     ELF headers
  ->     PT_INTERP      fallback   /glibc    /musl
  -> uspace.clear()
  -> map_trampoline()
  -> map_elf()     LOAD
  ->
```



```
  ->      auxv
load_user_app()
  ->
  ->      argv/envp/auxv
  ->          heap
```

ELF `LOAD` `Backend::new_cow`
trampoline `SIGNAL_TRAMPOLINE`

`kernel/src/task/user.rs::new_user_task`
`UserContext::run()` `PageFault(addr, flags)` mmap
`SIGBUS` `AddrSpace::handle_page_fault` `try_growdown_fault`
`SIGBUS` `SIGSEGV`

`AddrSpace::handle_page_fault` fault VMA
page size fault `populate`
fault

mmap brk

`sys_mmap` `MAP_PRIVATE` `MAP_SHARED` `MAP_SHARED_VALIDATE` `MAP_ANONYMOUS` `MAP_FIXED`
`MAP_FIXED_NOREPLACE` `MAP_GROWSDOWN` `MAP_LOCKED` `MAP_POPULATE`
CoW Shared File Linear Alloc mmap `/proc/pid/maps` growdown fault SIGBUS

`sys_brk` `ProcessData.heap_top` heap
heap Linux

`kernel/src/mm/access.rs` `UserPtr<T>`
`UserConstPtr<T>` `check_region`
`populate_area` null-terminated `access_user_memory`
`accessing_user_memory`

	`kernel/src/mm/aspace/mod.rs`	`AddrSpace` `va_range` `MemorySet<Backend>` `PageTable` map/unmap/protect/handle_page_fault/try_clone
	`kernel/src/mm/aspace/backend/mod.rs`	`Backend` `Linear/Cow/Shared/File` `MappingBackend`
ELF	`kernel/src/mm/loader.rs`	`map_elf` CoW `load_user_app` heap argv/envp/auxv
mmap/brk	`kernel/src/syscall/mm/mmap.rs`, `kernel/src/syscall/mm/brk.rs`	`sys_mmap` `sys_brk` `heap_top`
	`kernel/src/task/user.rs`	`ReturnReason::PageFault` `SIGBUS` `SIGSEGV`

StarryOS `axtask::TaskInner` `Thread` `TaskExt` `on_enter/on_leave` `scope-local` ArceOS `axtask` `kernel/Cargo.toml` `multitask` `task-ext` `sched-rr` feature StarryOS Linux

Thread

`Thread`

`proc_data`	`ProcessData`
`clear_child_tid`	`set_tid_address`
`robust_list_head`	robust futex `get_robust_list/set_robust_list`
`signal`	`ThreadSignalManager`
`time`	CPU interval timer
`exit` / `exit_event`	
`accessing_user_memory`	

`new_user_task` `UserContext::run()` syscall page fault interrupt exception `handle_syscall`

ProcessData

`ProcessData` `starry_process::Process`

`proc`	`starry_process::Process` pid /
`exe_path`, `cmdline`	`/proc` exec
`aspace`	
`scope`	scope-local
`heap_top`	`brk`
`rlim`	fd
`child_exit_event`, `exit_event`, `wait_status`, `stop_event`	wait job-control stop/continue
`signal`	`ProcessSignalManager` signal actions

`futex_table`	futex futex / key
`namespaces`	pid uts user namespace inode
`uid/euid/suid/gid/egid/sgid`	

/ `kernel/src/syscall/task` `clone/clone3` `execve` ELF `exit/exit_group` `wait` init `entry.rs` `Process::new_init(pid)` tty stdio

`task/resources.rs` `Rlimit` `Rlimits` `AX_FILE_LIMIT = 1024` `RLIMIT_STACK` `USER_STACK_SIZE` `RLIMIT_NOFILE` `AX_FILE_LIMIT` `ProcessNamespaces` pid uts user namespace inode `inherit_namespaces_from` clone flag namespace pid namespace init pid 1

	`kernel/src/task/mod.rs`	`Thread` `TaskExt` `AsThread` `TaskInner`
.	`kernel/src/task/user.rs`	`new_user_task` syscall page fault exception signal
	`kernel/src/task/mod.rs`	`ProcessData` scope rlimit signal futex namespace uid/gid
	`kernel/src/task/resources.rs`	`Rlimits::default` `RLIMIT_STACK` `RLIMIT_NOFILE`
. .	`kernel/src/syscall/task/*`	clone/execve/exit/wait/thread/job/schedule/ptrace

FileLike fd

`FileLike` trait socket eventfd timerfd signalfd epoll pidfd BPF map/program fd `FileLike` `read` `write` `stat` `ioctl` `poll` `path` `nonblocking` `status_flags` fd `scope_local` `Scope`

`AX_FILE_LIMIT` 1024 fd `FlattenObjects<FileDescriptor, AX_FILE_LIMIT>` `FileDescriptor` `cloexec` exec close-on-exec

`kernel/src/file/fs.rs` `axfs` `resolve_at` `AT_FDCWD` `AT_EMPTY_PATH` `AT_SYMLINK_NOFOLLOW` Linux `metadata_to_kstat` VFS metadata `Kstat` `From<Kstat> for stat/statx`

VFS node flags `poll_io` mount flags `O_NOATIME` atime `fcntl` I/O owner `ASYNC_FCNTL_STATE` pipe `SIGIO`

`pseudofs::mount_all` init

`/dev`	`devfs`	stub tty/pty loop random/zero/null/full rtc fb/input
`/dev/shm`	`tmpfs`	
`/tmp`	`tmpfs`	
`/proc`	`procfs`	sysctl
`/sys`	`tmpfs` +	graphics/fb0/device loop queue

`pseudofs/fs.rs` `SimpleFs` slab inode `pseudofs/tmp.rs` `MemoryFs` `pseudofs/device.rs` `DeviceOps` `DeviceMmap`

fd

`Pipe` ring buffer `PollSet` `SIGPIPE` `BrokenPipe` `pipe2` `O_CLOEXEC` `O_NONBLOCK`

`eventfd` `AtomicU64` semaphore poll `timerfd` deadline interval expiration worker `signalfd` pending signal mask Linux `signalfd_siginfo`

procfs

`kernel/src/pseudofs/proc.rs` `/proc/[pid]/task/[tid]` task status pagemap timens offsets dummy `DUMMY_MEMINFO` `DUMMY_CPUINFO` “ ”

`fd` . .	`kernel/src/file/mod.rs`	`FileLike` trait `FD_TABLE`
. .	`kernel/src/file/fs.rs`	`File`/`Directory` axfs `resolve_at` `metadata_to_kstat` Linux stat
	`kernel/src/pseudofs/mod.rs`	`mount_all` `/dev` `/dev/shm` `/tmp` `/proc` `/sys`
`tmpfs`	`kernel/src/pseudofs/tmp.rs`	`MemoryFs` inode symlink xattrs

`fd`	`kernel/src/file/pipe.rs`, `event.rs`, `timerfd.rs`, `signalfd.rs`	pipe/eventfd/timerfd/signalfd `FileLike` `Pollable`

framebuffer/input

GUI framebuffer/input
X11 Wayland

devfs

`kernel/src/pseudofs/dev/mod.rs` `/dev` `null` `zero` `full` `random` `urandom` `rtc0` `tty` `tty0` `console` `ptmx` `pts` `cpu_dma_latency` `vda` `vdd` stub `loop0` `loop15` `dev-log` `memtrack` feature

`DeviceOps` `read_at` `write_at` `ioctl` `as_pollable` `mmap` `len` `flags` VFS `Device` `FileNodeOps` `Pollable`
ioctl poll/epoll mmap

Framebuffer

`/dev/fb0` `axdisplay::has_display()` devfs `FrameBuffer::new` `axdisplay::framebuffer_info()` framebuffer `fb-refresh` 60Hz `axdisplay::framebuffer_flush()`

framebuffer

- `read_at` / `write_at` framebuffer
- `FBIOGET_VSCREENINFO` `FBIOPUT_VSCREENINFO` `FBIOGET_FSCREENINFO` `FBIOGETCMAP` `FBIOPUTCMAP` ioctl
- `mmap` `DeviceMmap::Physical` mmap linear

framebuffer X11/Wayland server

Input

`#[cfg(feature = "input")]` feature `/dev/input` `event::input_devices` `EventDev` `AxInputDevice` Linux evdev `input_event` `EVIOCGVERSION` `EVIOCGID` `EVIOCGRAB` `EVIOC*` ioctl `Pollable`

TTY PTY

TTY/PTY `kernel/src/pseudofs/dev/tty` `N_TTY` init `ptmx` pty master `pts` slave TTY ioctl termios / /job-control shell

`devfs`	`kernel/src/pseudofs/dev/mod.rs`	`builder` null/zero/random/tty/ptmx/loop/fb/input feature display

	`kernel/src/pseudofs/device.rs`	`DeviceOps` read/write/ioctl/poll/mmap `Device` VFS
`framebuffer`	`kernel/src/pseudofs/dev/fb.rs`	`FrameBuffer` framebuffer read/write/ioctl `DeviceMmap::Physical` mmap
`input`	`kernel/src/pseudofs/dev/event.rs`	`EventDev` `AxInputDevice` evdev `InputEvent`
`tty/pty`	`kernel/src/pseudofs/dev/tty/mod.rs`	`Tty` termios foreground pgrp winsize ptmx/pts poll

StarryOS `starry_signal` crate
`ProcessData` `Thread`

`ProcessData.signal`	`ProcessSignalManager`	pending signal signal actions trampoline
`Thread.signal`	`ThreadSignalManager`	pending/blocked signal manager
`ProcessData.wait_status`	`ProcessWaitStatus`	job-control stop/continue waitid
`ProcessData.stop_event`	`PollSet`	

trampoline `config::SIGNAL_TRAMPOLINE` ELF

`kernel/src/syscall/signal.rs` Linux

- `rt_sigaction` / signal action `SIGKILL` `SIGSTOP`
- `rt_sigprocmask` `SIG_BLOCK` `SIG_UNBLOCK` `SIG_SETMASK` blocked mask
- `rt_sigpending` pending signal
- `kill` pid 0 -1 pid Linux uid/euid/suid
- `tkill` / `tgkill` tid tgid+tid
- `sigsuspend` `sigaltstack` `rt_sigtimedwait`

`task/signal.rs::check_signals` ptrace stop
/ `SignalOSAction`

`Terminate`	`do_exit(signo, true)`

`CoreDump`	`signo | 0x80`
`Stop`	`wait_status = Stopped(signo)` continue
`Continue`	`wait_status = Continued(signo)` stop waiters
`Handler`	handler

Page fault `SIGBUS` `SIGSEGV` `SIGILL` `SIGTRAP` `SIGBUS`

signalfd

`signalfd4` syscall `file/signalfd.rs::Signalfd` `FileLike` `Pollable` mask pending signal 128 `signalfd_siginfo` poll/epoll

.	`kernel/src/syscall/signal.rs`	`rt_sigaction` `rt_sigprocmask` `kill/tkill/tgkill` signal manager
	`kernel/src/task/signal.rs`	`check_signals` `SignalOSAction` terminate/core/stop/continue/handler
.	`kernel/src/task/user.rs`	`check_signals` / fatal signal
`signalfd`	`kernel/src/syscall/fs/signalfd.rs`, `kernel/src/file/signalfd.rs`	`signalfd4` `Signalfd` read `signalfd_siginfo`

Feature

`starry-kernel/net` feature `qemu` feature `starry-kernel/net` QEMU `net` `kernel/src/syscall/net/mod.rs` socket stub `ENOSYS`

`net` `syscall/net` re-export `socket` `io` `name` `opt` `cmsg` `axnet` socket

Socket

`kernel/src/file/net.rs::Socket` `axnet::Socket` `FileLike`

- `read` `recv` `write` `send`
- `stat` socket `Kstat`
- nonblocking axnet socket option
- `poll`/`register` socket
- `path` `socket:[addr]`

fd eventfd fd poll/epoll

socket

`sys_socket`

domain	type	
`AF_INET` / `AF_INET6`	`SOCK_STREAM`	`axnet::tcp::TcpSocket` 0 TCP SCTP
`AF_INET` / `AF_INET6`	`SOCK_DGRAM`	`axnet::udp::UdpSocket` 0 UDP UDPLITE
`AF_UNIX`	`SOCK_STREAM` / `SOCK_SEQPACKET`	`UnixSocket` + `StreamTransport`
`AF_UNIX`	`SOCK_DGRAM`	`UnixSocket` + `DgramTransport`
`AF_VSOCK`	`SOCK_STREAM`	`vsock` cfg

`socketpair` `AF_UNIX` stream dgram seqpacket `bind` euid root 1 1023 `EACCES`

I/O

`sendto/sendmsg` `VmBytes` `IoVectorBuf` control message `recvfrom/recvmsg` `MSG_PEEK` `MSG_TRUNC` `SCM_RIGHTS` `getsockopt/setsockopt` `SO_REUSEADDR` `SO_ERROR` `SO_SNDBUF` `SO_RCVBUF` `SO_KEEPALIVE` `SO_RCVTIMEO` `SO_SNDTIMEO` `SO_PASSCRED` `SO_PEERCRED` `TCP_NODELAY` `TCP_MAXSEG` `TCP_INFO` `IP_TTL`

`setsockopt(SO_ATTACH_BPF)` fd `BpfProgram` socket option BPF

.	`kernel/src/syscall/net/mod.rs`	`net` socket syscall `ENOSYS` socket/io/name/opt/cmsg
`socket`	`kernel/src/syscall/net/socket.rs`	`sys_socket` domain/type TcpSocket UdpSocket UnixSocket
`socket fd`	`kernel/src/file/net.rs`	`Socket` `FileLike` `Pollable` axnet send/recv/poll
. `I/O`	`kernel/src/syscall/net/io.rs`	`sendmsg/recvmsg` iovec control message
`socket options`	`kernel/src/syscall/net/opt.rs`	`getsockopt/setsockopt` SOL_SOCKET/TCP/IP `SO_ATTACH_BPF`

IPC

Futex

`kernel/src/task/futex.rs` futex wait queue futex table `FutexKey` shared mmap futex `SharedPages`

futex handle

wait queue

`sys_futex` `FUTEX_WAIT` `FUTEX_WAIT_BITSET` `FUTEX_WAKE` `FUTEX_WAKE_BITSET` `FUTEX_REQUEUE` `FUTEX_CMP_REQUEUE` fast path `WaitQueue::wait_if` bitset robust list `sys_get_robust_list` `sys_set_robust_list` `Thread.robust_list_head`

SysV IPC

`syscall/ipc` message queue shared memory

Message Queue	`syscall/ipc/msg.rs`	`MessageQueue` `msqid_ds` mtype `MsgManager` key msqid msqid
Shared Memory	`syscall/ipc/shm.rs`	`ShmInner` shmid attach `ShmidDs` `ShmManager` key/shmid/pid/vaddr

`SharedPages` `Backend::Shared` mmap shared backend

I/O

`epoll` syscall `sys_epoll_create1` `Epoll` fd `sys_epoll_ctl` / / interest `sys_epoll_pwait` `sys_epoll_pwait2` `poll_io` ready events `file/epoll.rs` interest level-trigger edge-trigger one-shot Waker fd readiness ready queue

`poll` `ppoll` `select` `pselect` `syscall/io_mpx` `FileLike::poll/register` readiness

eventfd pipe signalfd

`FileLike + Pollable`

- `eventfd` fd
- `pipe` ring buffer `FIONREAD` `SIGPIPE`
- `signalfd` pending signal poll fd
- `timerfd` expiration fd

/

`futex`	`kernel/src/task/futex.rs`	`FutexKey` private/shared/file `WaitQueue` wait/wake/requeue

`futex syscall`	`kernel/src/syscall/sync/futex.rs`	`sys_futex` wait/wake/requeue robust list
	`kernel/src/syscall/ipc/msg.rs`	`MessageQueue` `MsgManager` SysV message queue
	`kernel/src/syscall/ipc/shm.rs`	`ShmInner` `ShmManager` shared pages attach/detach shmid
`epoll`	`kernel/src/syscall/io_mpx/epoll.rs`, `kernel/src/file/epoll.rs`	`Epoll` fd interest trigger mode ready queue

BusyBox shell `CMDLINE` `/musl/busybox sh -c` `INIT_SCRIPT` `entry.rs` BusyBox `/musl` `/glibc` `/bin` fallback `loader.rs` `/glibc` `/musl` fallback musl/glibc rootfs

`src/init.sh` rootfs LoongArch RISC-V glibc loader `/lib` musl/glibc runtime basic busybox iozone libcbench lmbench ltp lua netperf

`src/init.sh` shell

`iozone=1`	iozone /
`libcbench=1`	libcbench
`lmbench=1`	lmbench
`lua=1`	lua
`netperf=1`	netperf
`basic/busybox/cyclictest/iperf/libctest/ltp=0`	`some_testcases`

`src/main.rs` `src/ltp_testcases.sh` `LTP_CASES` `init.sh` `run_ltp` LTP Summary TPASS/TFAIL/TBROK/TCONF/TWARN summary

rootfs

`init.sh` wrapper

- `/etc/passwd` `/etc/group` `/etc/resolv.conf`
- `/lib/modules/10.0.0/build/.config` `CONFIG_*`
- `useradd/userdel/groupdel`

- `mkfs.ext3` `mkfs.ext4` wrapper no-op
- `/bin` BusyBox

`BusyBox init`	`src/main.rs`	`CMDLINE` `/musl/busybox sh -c INIT_SCRIPT` feature init
`fallback`	`kernel/src/entry.rs`, `kernel/src/mm/loader.rs`	init musl/glibc/bin BusyBox fallback `/glibc` `/musl`
	`src/init.sh`	rootfs LTP summary
`LTP` .	`src/main.rs`, `src/ltp_testcases.sh`	`build_ltp_cases_env` `LTP_CASES`

init

`src/init.sh`	rootfs benchmark/LTP
`src/init_shell.sh`	`shell-init` feature shell
`src/ltp_testcases.sh`	LTP `src/main.rs` `LTP_CASES`
`syscalls_code/`	
`docs/typst_doc`	Typst

rootfs

init.sh

`src/init.sh` `#### OS COMP TEST GROUP START ltp-$1 ####`

`run_ltp` LTP summary Summary

`TPASS` `TFAIL` `TBROK` `TCONF` `TWARN`

QEMU group marker

1. `main.typ` `#include`
2. `typst compile main.typ build/main.pdf` PDF
3.

4. “ ”
5. “ ”“X11/Wayland”“ ”

	`src/init.sh`	basic/busybox/iozone/libcbench/lmbench/ltp/ lua/netperf
`LTP` .	`src/main.rs`	`build_ltp_cases_env` `src/ltp_testcases.sh` `LTP_CASES`
.	`syscalls_code/`	

StarryOS ArceOS/ax Linux
Rust
`FileLike` `AddrSpace` `ProcessData`

ELF / mmap/brk CoW shared/file/linear
`axtask` Linux axfs
`/dev` `/proc` tmpfs pipe eventfd timerfd signalfd socket epoll Linux fd

- `/proc/meminfo` dummy
- `/lib/modules/.../.config`
- framebuffer input GUI X11/Wayland
- BPF map/prog fd eBPF verifier/JIT/runtime
- socket `net` feature `axnet` feature `ENOSYS`

1. / rootfs QEMU
2. summary
3. OS

. .	`kernel/src/lib.rs`	entry config/file/mm/pseudofs/ syscall/task/time
. .	`kernel/src/syscall/mod.rs`	`handle_syscall` Linux syscall

	`kernel/src/entry.rs`	init unmount rootfs flush system_off

A

Cargo workspace

`Cargo.toml` workspace `kernel` package `starryos` `src/main.rs` `[[bin]]` `qemu` feature QEMU `--features qemu` /xtask

feature

Feature	
`qemu`	`axfeat/defplat` `axfeat/bus-pci` `starry-kernel/net`
`shell-init`	`src/init_shell.sh` `STARRY_SHELL_MODE=1`
`timerfd-debug-init`	`src/main.rs` timerfd init feature
`smp`	`axfeat/smp`
`vf2`	`axfeat/driver-sdmmc` `src/main.rs` VisionFive2 crate

crate feature

Feature	
`net`	`axfeat/net` `axnet`
`input`	`axinput` `axfeat/input` `/dev/input`
`dev-log`	`/dev/log`
`memtrack`	axalloc tracking dwarf/backtrace `/dev/memtrack`

Typst `docs/typst_doc/main.typ`

```
typst compile main.typ build/main.pdf
```

`docs/typst_doc/build/main.pdf` LaTeX

`workspace`	`Cargo.toml`	`members = ["kernel"]` `[[bin]] name = "starryos"` required feature `qemu`
. . `feature`	`kernel/Cargo.toml`	`net/input/dev-log/memtrack`
· · ·	`docs/typst_doc/main.typ`	Typst `#include` `template.typ`

B

syscall kernel/src/syscall/mod.rs handle_syscall match

	fs/ctl.rs	chdir mkdirat unlinkat renameat2 sync xattr
fd I/O	fs/fd_ops.rs, fs/io.rs	openat close dup3 fcntl read/write pread/pwrite sendfile
fd	fs/pipe.rs, fs/event.rs, fs/signalfd.rs, fs/pidfd.rs	pipe2 eventfd2 signalfd4 pidfd_*
I/O	io_mpx/*	epoll_create1 epoll_ctl epoll_pwait ppoll pselect
	mm/*	brk mmap munmap mprotect mremap mincore
	task/*	clone clone3 execve wait4 exit_group sched_yield ptrace
	signal.rs	rt_sigaction rt_sigprocmask kill tgkill sigsuspend
	net/*	socket bind connect accept4 sendmsg recvmsg getsockopt
IPC/	ipc/*, sync/*	msgget/msgsnd/msgrcv shmget/shmat futex membarrier
/ /	time.rs, resources.rs, sys.rs	clock_gettime nanosleep getrlimit/prlimit uname
BPF	bpf.rs	bpf map/prog

	kernel/src/syscall/mod.rs	handle_syscall
.	kernel/src/syscall/*	mod fs/mm/task/net/ipc/sync/io_mpx/resources/signal/time/sys/bpf

C

AddrSpace	kernel/src/mm/aspace/mod.rs	VMA

`Backend`	`kernel/src/mm/aspace/backend/mod.rs`	Linear Cow Shared File
`Thread`	`kernel/src/task/mod.rs`	signal timer robust list
`ProcessData`	`kernel/src/task/mod.rs`	Process scope rlimit signal futex namespace uid/gid
`ProcessNamespaces`	`kernel/src/task/mod.rs`	pid/uts/user namespace inode
`FileLike`	`kernel/src/file/mod.rs`	trait read/write/stat/ioctl/poll/path/status flags
`Kstat`	`kernel/src/file/mod.rs`	stat Linux `stat` `statx`
`DeviceOps`	`kernel/src/pseudofs/device.rs`	read/write/ioctl/poll/mmap
`FutexKey` / `FutexTable`	`kernel/src/task/futex.rs`	futex wait queue
`MessageQueue`	`kernel/src/syscall/ipc/msg.rs`	SysV message queue
`ShmInner`	`kernel/src/syscall/ipc/shm.rs`	SysV shared memory attach `ShmidDs`

	`kernel/src/mm`, `kernel/src/task`, `kernel/src/file`, `kernel/src/pseudofs`, `kernel/src/syscall/ipc`	

D

	2026-06-09
	git commit dirty
	riscv64 loongarch64 x86_64 aarch64
	cargo/xtask/qemu feature
rootfs	musl/glibc
	basic busybox ltp iozone lmbench netperf

	/QEMU/
	pass/fail/skip

`src/init.sh`	
`src/ltp_testcases.sh`	LTP
`src/init_shell.sh`	shell
`syscalls_code/`	
`docs/io_benchmark_pass_optimization.md`	I/O benchmark

	`src/init.sh`	`run_ltp`
	`src/main.rs`	`LTP_CASES_RAW` `src/ltp_testcases.sh`
	`syscalls_code/`	